operations CO-4

Machine Learning Operations

* MLOps is a set of engineering practices that unifies machine learning, DevOps and data engineering to automate, secure and scale the entire lifecycle.

→ Automates ML workflows.
→ Reduces manual effort.
→ Improves collaboration.
→ Enables continuous deployment

MLOps Architecture

Components

→ Data Sources
→ Data pipeline

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→ Feature Store
→ Model Training.
→ Model Registry
→ Deployment Platform
→ Monitoring System.

MLOps Principles

→ Version Control
→ Automation.
→ CI/CD
→ Testing & Validation
→ Monitoring.
→ Reproducibility.

MLOps Lifecycle

Data collection
↓
Data Preparation
↓
Model Development
↓
Model Training.
↓
Model Validation
↓
Deployment
↓
monitoring
↓
Retraining.

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Applications of MLOps

→ Health care = Disease prediction & diagnosis.
→ Retail = Product Recommendation
→ Finance = Fraud detection & risk assessment.
→ Manufacturing = Predictive maintenance.

MLOps Tools

→ ML Flow
→ Kubeflow
→ Tensor Flow Extended
→ Apache Airflow.
→ Jenkins
→ Docker.

Data Ingestion in MLOps

* Process of collecting, importing, validating and transferring data from various sources into a storage system or M.L pipeline

Data Ingestion Pipeline

Data Sources
↓
Validation
↓
Pre-processing
↓
Feature Engineering.

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Storage
↓
ML Training
↓
Deployment

Feature Engineering

* Process of creating meaningful features from raw data.

→ Feature Extraction
→ Feature Selection.
→ Feature Transformation.
→ Dimensionality Reduction.

Model Training

* Process of teaching a M.L model using data.

Model Development

*Model Development is the process of creating a M.L solution for a problem.

Training pipeline workflow

Raw Data
↓
Preprocessing
↓
Feature Engineering
↓
Training
↓

Validation

↓

Testing

↓

Deployment

Hyperparameter Tuning Techniques

1) Grid Search
2) Random Search
3) Bayesian optimization.

Model Evaluation

* Process of measuring how well a trained M.L model perform

Data Version Control

* Open-source tool for machine learning & data science projects.
* Works with Git.
* Helps to manage & track datasets, models, and experiments.

→ Data Versioning.
→ Model Versioning.
→ Pipeline Tracking
→ Git Integration
→ Experiment Reproducibility.

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Data Versioning

* Tracks & manages changes made to datasets over time

Model Versioning

* Process of tracking & managing different versions of M.L.

Pipeline Tracking

* Process of monitoring & managing each stage of the M.L workflow.

MLFlow

* Open-source platform designed to manage the complete M.L Lifecycle.

Lifecycle

Experiment Tracking
↓
Model Registry
↓
Model Packaging
↓
Model Deployment
↓
Serving Models.

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Monitoring

* Continuous observation of syst applications & infrastructure.

Prometheus

* Open-source monitoring & alerti toolkit used to collect, store, analyze metrics from applicatio & infrastructure.
* Stores metrics as time-series data.
  - → PromQL (Powerful Query Language)
  - → Alerting support.
* Metrics Collection
* Alerting.
* Model Drifting Detection.
  - → Data Drift
  - → Concept Drift
  - → Prediction Drift.
* Performance Monitoring.

AI Bias

* Refers to systematic & unfair favoritism or discrimination in AI system outcomes.

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Types of Bias in AI

1.) Sampling Bias => Non-representative data.

2) Historical Bias => Existing societal inequalities.

3.) Confirmation Bias => Reinforcing existing assumptions.

4.) Labeling Bias => Human errors during annotation

5.) Automation Bias => Over-reliance on AI decisions

AI Ethics

* Moral principles & guidelines that govern the design, development, deployment.

Key Ethical Principles

→ Fairness
→ Transparency.
→ Accountability.
→ Privacy
→ Safety & Security.

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Cloud-Native MLOps

* Process of building, training, deploying & monitoring M.L models using cloud services.

→ Scalable resources.
→ Automated setup.
→ Distributed systems.
→ Easy scaling.

AWS ⇒ AWS Sagemaker
Azure ⇒ Azure ML Studio.
GCS ⇒ Vertex AI.

CI/CD Workflow

Code Commit
↓
Validation
↓
Training
↓
Testing
↓
Packaging
↓
Deployment

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